

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Koppala Navya	Reg. No.:	192511231
Section / Batch:	B-E CSE	Date:	07/08/2026

Code of Conduct

I, Koppala Navya [192511231] (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Navya

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Question - 01

A hospital treats patients on a First-Come, First-Served (FCFS) basis. Patients arrive in the following order

P₁ (General)

P₂ (Emergency)

P₃ (General)

P₄ (General)

P₅ (Emergency)

(a) Treatment order using a normal queue

⇒ A normal queue follows the First-Come, First-Served (FCFS) principle. This means the patient who arrives first is treated first, regardless of whether the patient is a general or emergency case

∴ The treatment order will be

$P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_4 \rightarrow P_5$

* Here P₁ is treated first because P₁ arrived first. P₂ is although an emergency patient must wait until P₁ is treated.

(b) why is a normal queue not suitable?

⇒ A normal FCFS queue is not suitable for a hospital because it gives equal importance

to all patients and does not consider the seriousness of their medical condition. Hence

⇒ For example, in this case, P2 is an emergency patient, but P2 has to wait behind P1, who is a general patient similarly, P5 is also an emergency patient but would have to wait behind P3 and P4.

⇒ In a real hospital, delaying an emergency patient, but P2 has to wait behind P1, who may put the patient's life at risk.

⇒ Therefore, a queue system that considers priority or urgency is more appropriate.

(c) Better Queue-based solution:

⇒ A priority queue is the better solution.

⇒ In a priority queue, every patient is assigned a priority based on their medical condition.

* Emergency patients → High priority

* General patients → Low priority.

Therefore:

* P2 and P5 are emergency patients, so they are treated first.

* P2 arrived before P5, so P2 is treated first

* P1, P3 and P4 are general patients

* Among them treatment order becomes.

$$P2 \rightarrow P5 \rightarrow P1 \rightarrow P3 \rightarrow P4$$

Conclusion:

A priority queue is more suitable for a hospital because it ensures that patients with serious or emergency conditions receive treatment before patients with less serious conditions. It also maintains fairness by using FCFS for patients having the same priority.

Question 02:

A calculator application receives the following expression:

$$(10+5) * (3-4) / 2$$

The given expression is :

$$(10+5) \times (3-4) / 2$$

To solve this expression, we follow the order of operations. First, we solve the expressions inside the brackets.

Step 1: solve the first bracket

$$10+5 = 15$$

Step 2: solve the second bracket

$$3-4 = -1$$

Now the expression becomes:

$$15 \times (-1) / 2$$

Step 3: Perform Multiplication

$$15 \times (-1) = -15$$

So:

$$-15 / 2$$

Step 4: perform division

$$-15 \div 2 = -7.5$$

Final Answer

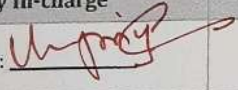
$$(10+5) \times (3-4) / 2 = -7.5$$

Therefore, the correct answer to Question 2 is

$$-7.5$$

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:		100 Marks	

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____